

One valid criticism of CPPIB's factor investing procedure is that it doesn't go far enough! The factors in the Reference Portfolio are simple: static equities and bonds. There are no dynamic factors in CPPIB's benchmark, at least not yet.

3.4. HOW ARE YOU DIFFERENT FROM AVERAGE?

How do we allocate to factors, especially dynamic factors?

The skeptics will facetiously reply that we can allocate to dynamic factors just as badly as we do regular asset allocation. They have a point. Industry is besotted with mean-variance analysis (see chapter 3), which is great for those rare investors with mean-variance preferences. (I don't know any, do you?) In mean-variance land, you still have to assign means and variances, along with correlations, to all the assets, and there is not much difference between using factors and regular asset classes.²² This is really a statement about the shortcomings of mean-variance optimization.

As Part I of this book stresses, modeling the investor is about identifying her bad times, and how well she can handle losses when losses materialize in bad times. We capture these notions with *utility functions*: a rich array defines bad times as low wealth, when your consumption falls below what you're used to, and even when your neighbor bests the performance of your own portfolio (see chapter 1). Factors perform badly during bad times, as they did for Norway in 2008 and 2009. The key in factor investing is comparing how the bad times encapsulated in factors compare to the bad times of an investor. Each investor will have a different set of bad times defined by her liabilities, income stream, how she tolerates losses (or not), and other salient investor characteristics.²³

We can, and should, use all the impressive machinery of utility functions and fancy optimization in allocating to factors. I am, however, not a big fan of blindly applied brute-force mathematical procedures. I now lay out a procedure for thinking about factor investing that focuses on an individual's bad times. We start with a very special investor who actually takes on no dynamic factor risk whatsoever . . .

Mr. Market

The market is the quintessential typical investor—the market, by definition, embodies the average effect. *The average investor holds the market portfolio*, a passive, long-only collection of all available securities held with market capitalization

²² As Kaya, Lee, and Wan (2011) and Idzorek and Kowara (2013) claim. Even in a mean-variance factor context, using factors does shrink, considerably, the number of means, variances, and correlations that need to be estimated, so there are some important differences between unconstrained approaches and factor models. See chapter 3.

²³ See Cochrane (1999, 2013a).